

ECE 4530/5530 Lab 4 Report

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Topology

From our specs, we want to achieve a high gain but a small output resistance. To drive such a small resistive load, we needed to use a two-stage OTA. For our topology, we initially started our design with the two-stage telescopic OTA because of its high gain. We considered the use of a folded cascode to improve input and output swing, but since we had a stringent gain requirement, we opted for the standard telescopic design. With aggressive biasing, we were able to achieve a gain only slightly better than 70dB. We needed to improve our gain by several orders of magnitude. We considered 3 possibilities:

- Adding a third gain stage
- Triple cascoding the input stage
- Cascoding the second stage

The first design was eliminated because of the instability and would require nested Miller compensation. The second design was chosen for a short while, but was quickly dropped because of the poor voltage headroom and difficulty in biasing 2 cascoded structures in a single stage. We opted for the second stage. We realized that the second stage would provide us with sufficient gain; it had a really high output impedance because of the cascode. We need really large output stage transistors to have an output resistance of 10kOhms. We also had to add additional bias resistors to properly bias the output stage cascode. After reaching our gain spec, we also needed to add Miller compensation to the op amp to meet the phase margin specs. This resulted in a severe hit to our unity gain bandwidth, but not enough to fail specs. Some other tweaks were necessary to increase the length of the transistors in the bias network, and the first stage increases the output impedance of the current mirrors.

Design Equations

First stage gain: $Av_1 = (g_m r_o)^2 / 2$

Second stage: $Av_2 = (g_m r_o)^2 / 2$

Unity Gain Bandwidth: $BW = |A(j\omega_{180})B(j\omega_{180})| > 1$ at 0dB

Phase Margin: $PM = \varphi(\omega_{UGBW}) + 180^\circ$

Schematic

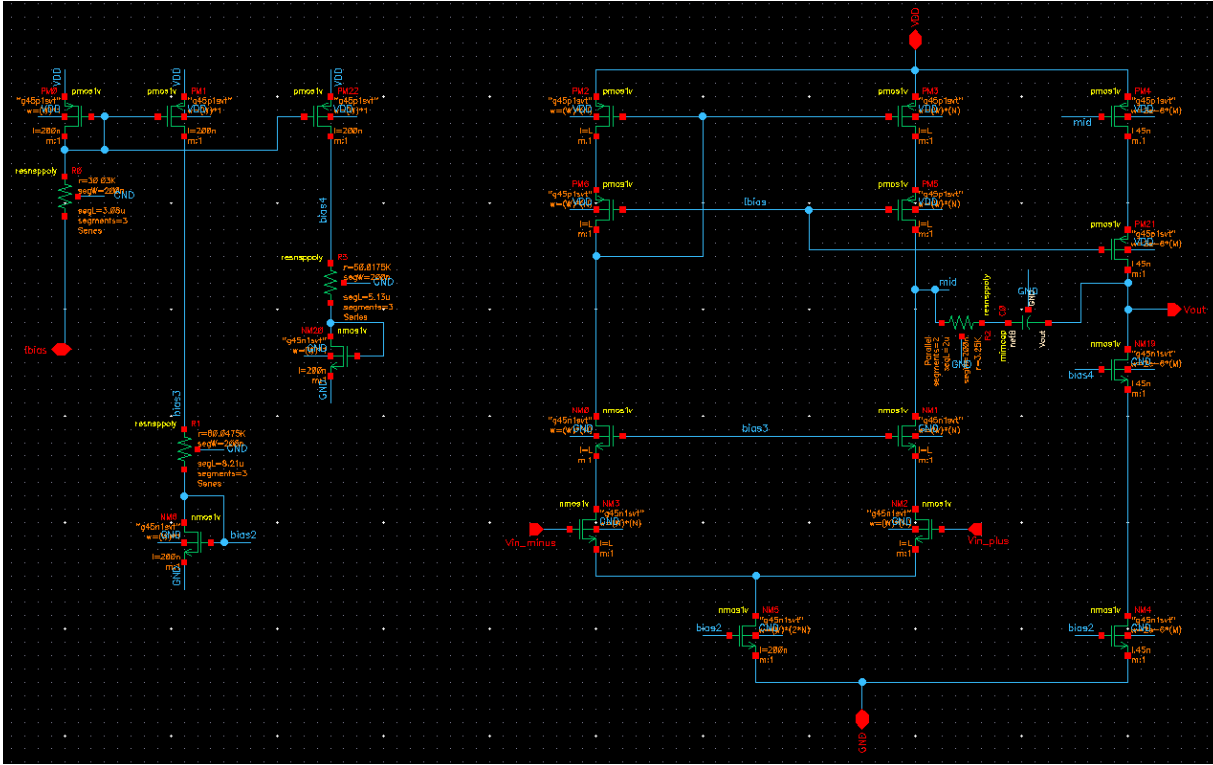


Figure 1: Op-Amp Schematic

Transistor Sizes

Instance Names	Length (L)	Width (W)
PM0, PM1, PM22, NM6, NM20, NM5	200nm	1μm
NM3, NM2, NM0, NM1, PM2, PM3, PM6, PM5	200nm	0.5μm
NM4, NM19, PM4, PM21	45nm	60μm

Table 1: Transistor Sizing

Specifications

Performance metric	Specifications	Achieved	Difference
Open loop gain V_{out}/V_{ind}	90dB	91.3055dB	1.3055dB
UGBW	25MHz	272.374MHz	247.374MHz
Phase margin	60°	61.955°	1.955°

Output range	1.0V	1.57752V	0.57752V
P_{DC}	10mW	2.86511mW	7.13489mW
R_{out}	10k Ω	9.23065k Ω	0.76935k Ω

Table 2: Design Specifications

What to measure	What you measured	Units
CMRR (open loop)	70.245	dB
PSRR (open loop)	80.525	dB
$I_{out\ max}$	4.45699m	A
$I_{out\ min}$	-824.915 μ	A
Slew rate (in Gain = 10FB)	20.54677M	V/s
Max cap load (in Gain= 10 FB)	150	pF
Max output range (in Gain = 10)	2.4317	V

Table 3: Miscellaneous Specifications

Our C_L did not cause instability for values in the pF range, so we plotted the transient for larger and larger values of C_L . We swept values of C_L up to 1kF. We chose 150pF as appropriate C_L because the transient curve at that curve experiences significant underdamped oscillations.

DC Simulation

Results Display Window	
Window Expressions Info Help	cadence
signal	OP("/V1" "??")
i	-1.14604m
pwr	-2.86511m
v	2.5

Figure 3: DC Power Consumption Report

Our DC power consumption is **2.86511mW**.

AC Simulations

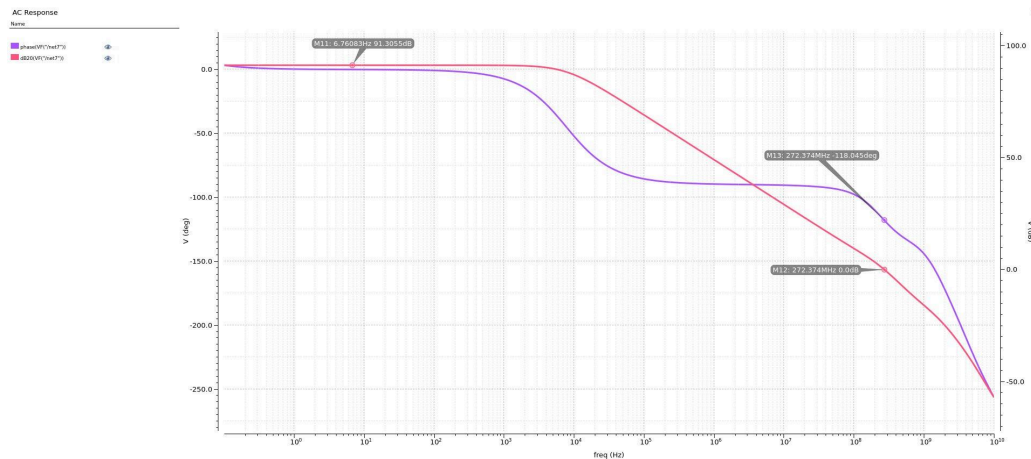


Figure 4: AC Simulation for Open Loop Gain, UGBW, and PM

We found that our gain was **91.3055dB**. Our UGBW was determined to be **272.374MHz**, which is the frequency in which magnitude is 0dB. We can calculate PM by adding 180° to the phase at the UGBW, which turned out to be **61.955°**.

To find our output impedance, we injected an AC current onto the output of the amplifier and set our input voltage to have 0 AC magnitude. We then measured the output voltage.

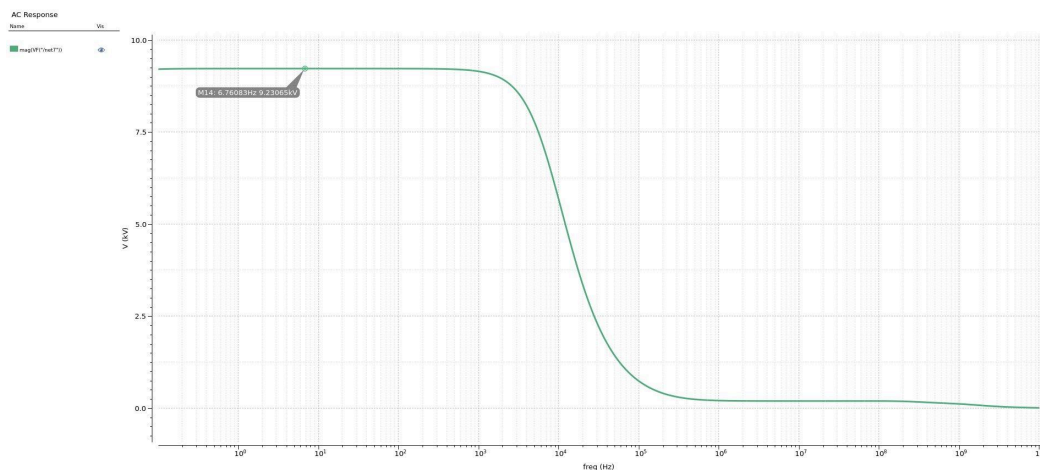


Figure 5: AC Magnitude with AC Output Current & DC Input Voltage

We found our R_{out} to be **9.23065k Ω** .

Transient Simulation

For calculating our output voltage swing, we ran a transient simulation, incrementally increasing the amplitude of our input signal until the transient gain dropped by about 3dB.

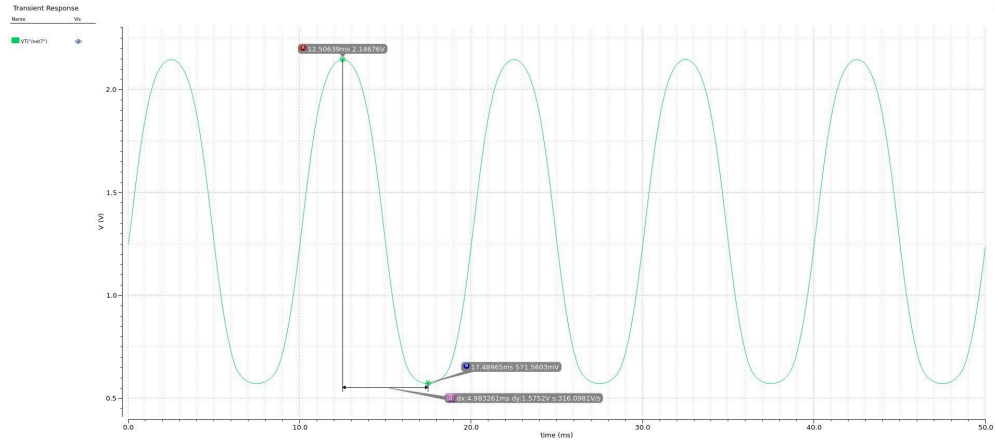


Figure 6: Open-Loop Transient Output

With an input amplitude of $30\mu\text{V}$, we found our output voltage range to be **1.57752V**.

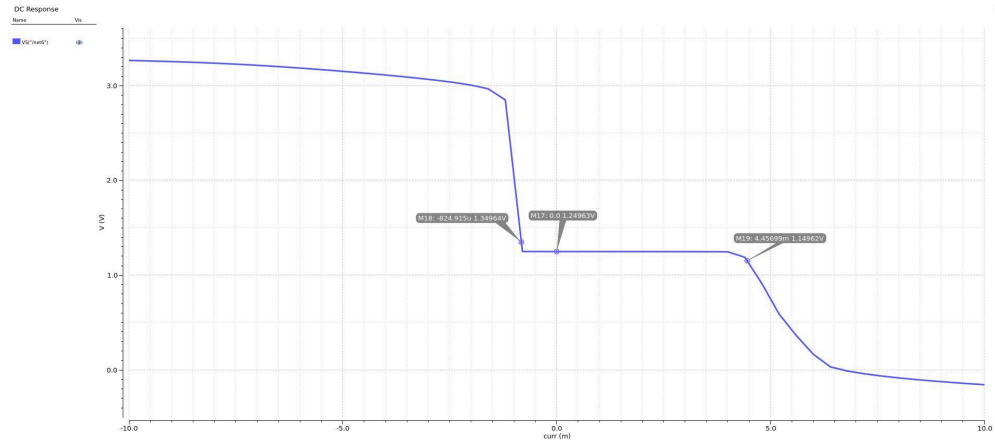


Figure 7: DC Output Current Sweep

The minimum output current was **-824.915 μA** and our maximum output current was **4.45699mA**, both of which were found where the output voltage changed by 100mV.

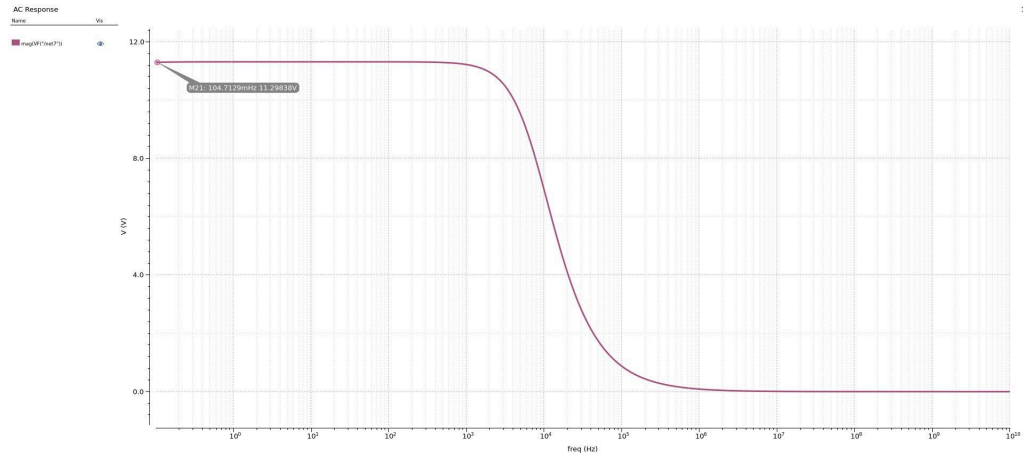


Figure 8: AC Simulation for Common Mode Gain

$$\text{CMRR} = \frac{36751.49413}{11.29838} = 3252.810945 = \mathbf{70.245\text{dB}}$$

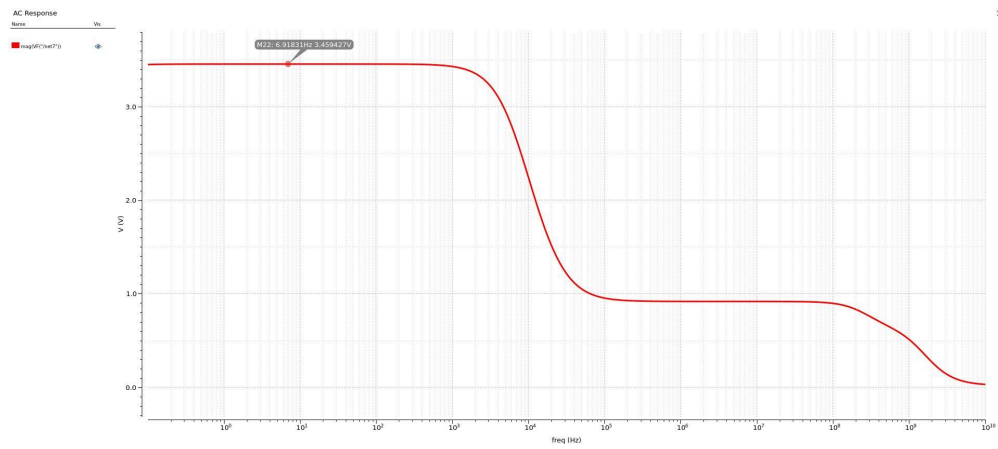


Figure 9: AC Simulation for PSRR

$$\text{PSRR} = \frac{36751.49413}{3.459427} = 10623.57845 = \mathbf{80.5254\text{dB}}$$

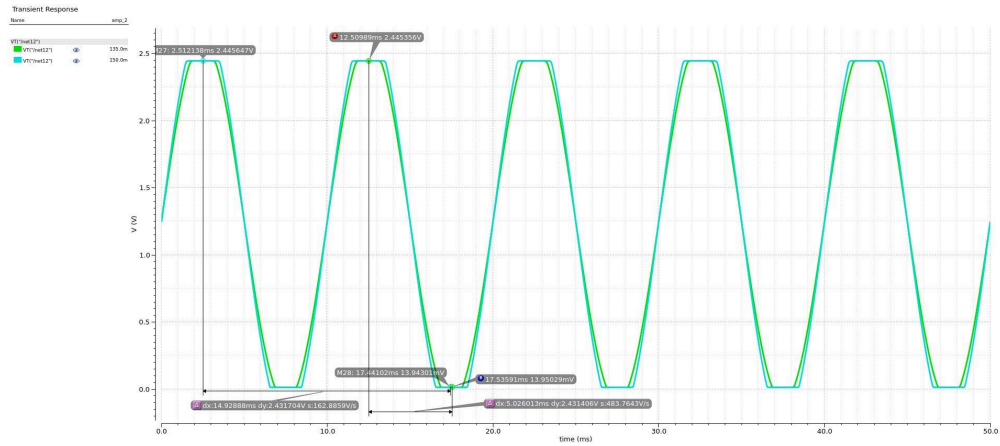


Figure 10: Closed-Loop Transient Output

With an input amplitude of 150mV, our output range was **2.4317V**.

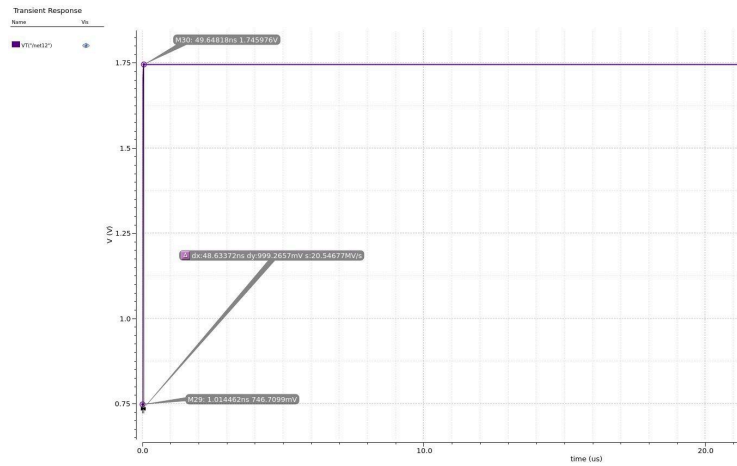


Figure 11: Output Response of Step Input

With a 100mV step on the input, our slew rate was calculated to be **20.54677MV/s**.

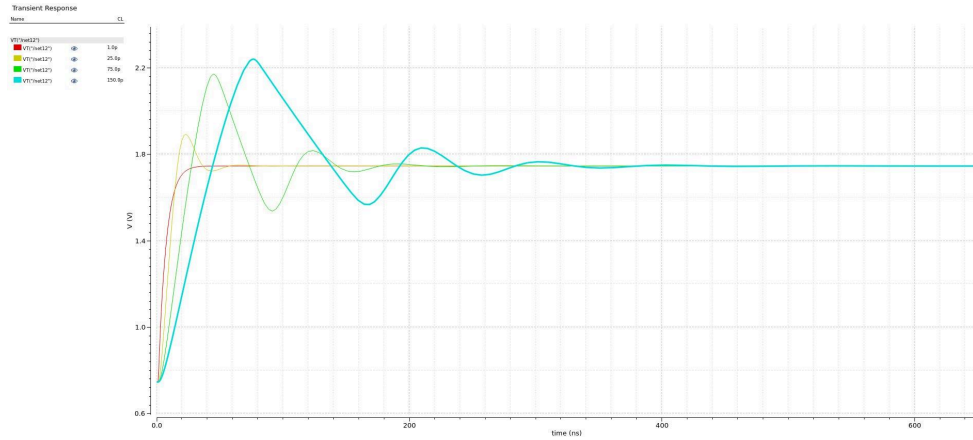


Figure 12: C_L Transient Sweep

Our maximum C_L load was found to be around **150pF**.

Conclusion

In conclusion, we have designed an op amp that meets the desired specifications after theoretical analysis and rigorous testing. Our amplifier demonstrates an open loop differential gain of 91.3055 dB (greater than 90 dB by 1.3055 dB), unity gain bandwidth of 272 MHz (over 25 MHz by 247.374MHz), phase margin of 61.955° (greater than 60° by 1.955°), power consumption of 2.86511 mW (less than 10 mW by 7.13489 mW), output swing of 1.57752 V peak-to-peak (greater than 1 V by 0.57752 V), and small signal output resistance of 9.23065 k Ω (less than 10 k Ω by 0.76935 k Ω). This was achieved using a two-stage OTA with a cascoded second stage and Miller compensation.

Layout

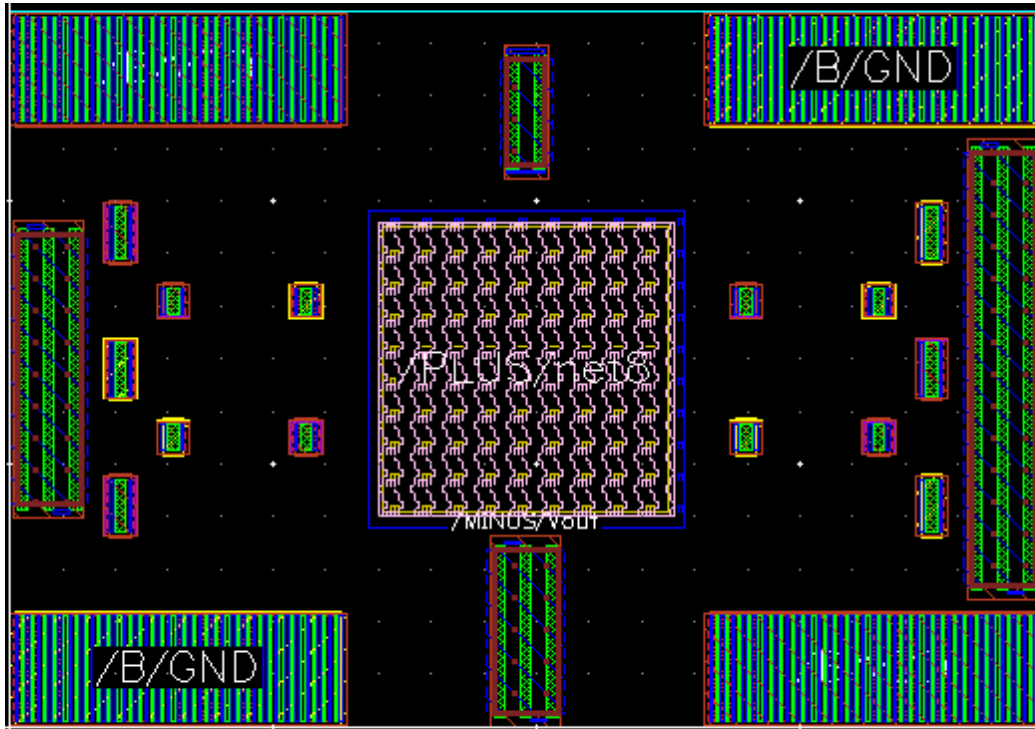


Figure 13: Components in Common Centroid

Our circuit was placed using a common centroid scheme, where all matched components share the same mean location. We had three types of transistors: 8 with $W = 0.5\mu\text{m}$ & $L = 200\text{nm}$, 6 with $W = 1\mu\text{m}$ & $L = 200\text{nm}$, and 4 with $60\mu\text{m}$ & $L = 45\text{nm}$. Our compensation capacitor is one MIMCAP, and was placed in the middle.

Since our resistors were all different sizes, they were placed to ensure that for each resistor, its center had the same distance to the overall center as the resistor across from it. The larger resistors were placed opposite each other on the left and right edges, and the smaller resistors were placed opposite each other on the top and bottom edges.

To determine placement of certain transistors, its connecting components were considered, such as placing components connected to V_{DD} closer to the top, keeping groups of components together, etc. This, however, was not a strict rule.

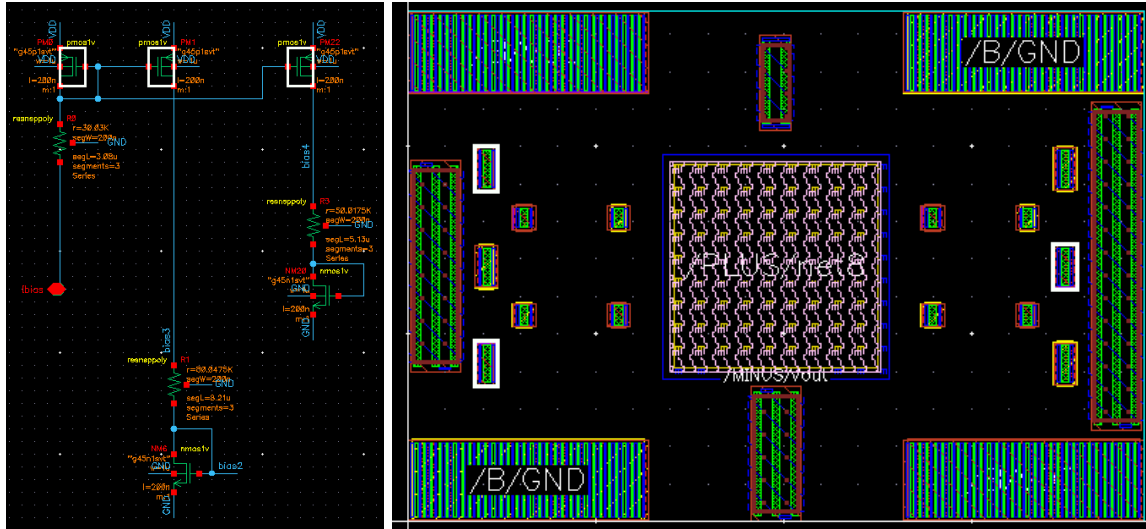


Figure 14: PMOS Transistors (Biasing Circuit) Schematic and Layout

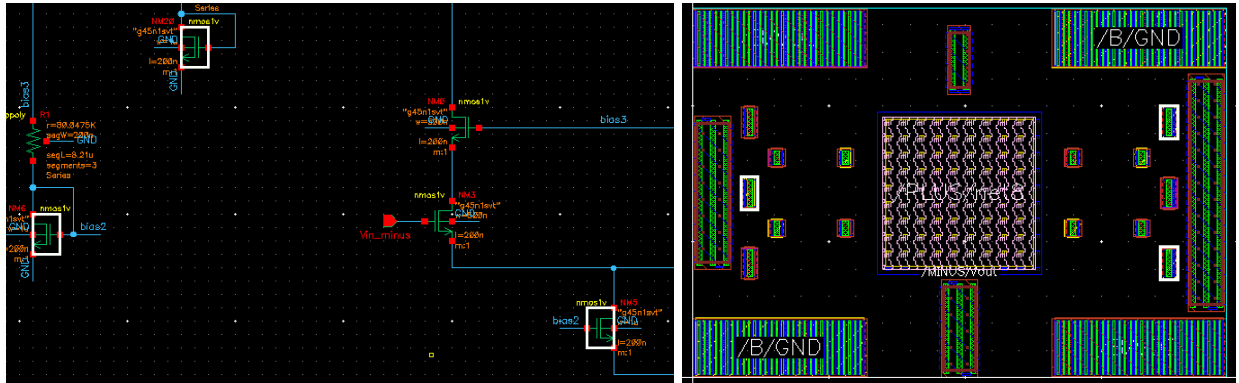


Figure 15: NMOS Transistors (Biasing Circuit) Schematic and Layout

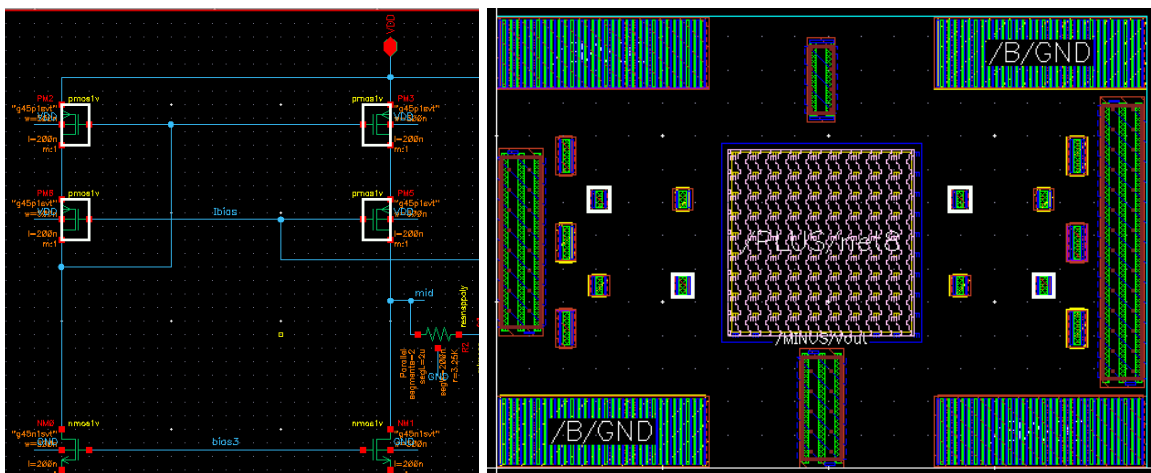


Figure 16: PMOS Transistors (Folded Cascode) Schematic and Layout

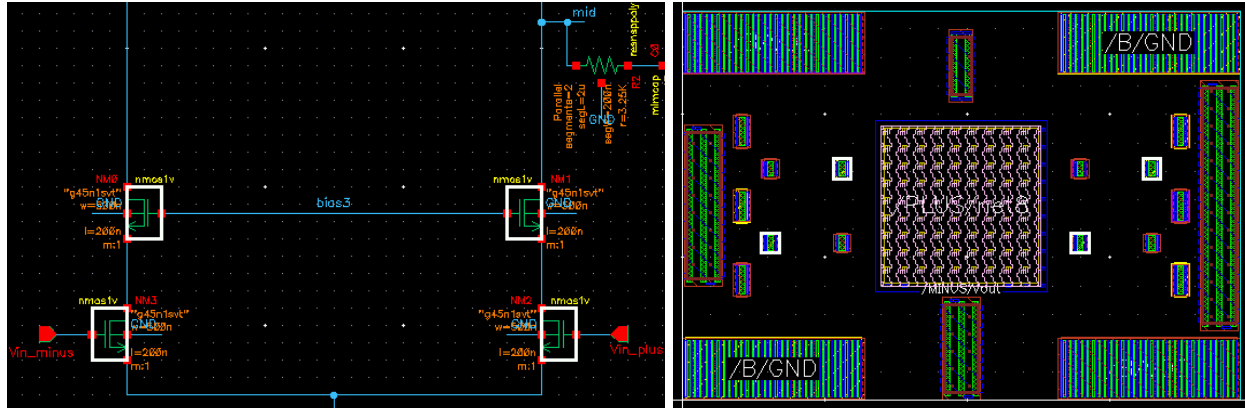


Figure 17: NMOS Transistors (Folded Cascode) Schematic and Layout

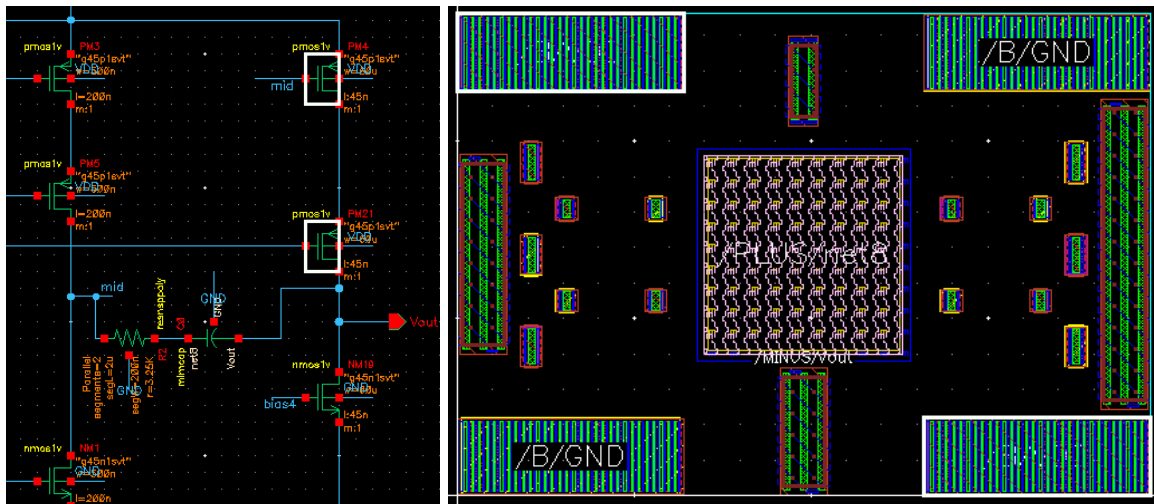


Figure 18: PMOS Transistors (Cascode Output) Schematic and Layout